MIL-PRF-19500/452E
13 May 1999
SUPERSEDING
MIL-S-19500/452D
1 October 1997

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, LOW LEVEL VOLTAGE-REFERENCE, TYPES 1N4565A-1 THROUGH 1N4584A-1, AND 1N4565AUR-1 THROUGH 1N4584AUR-1 JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC; RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H; JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; AND JANKCM, D, L, R, F, G, H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 6.4 volts ±5 percent, silicon, low bias current, voltage-reference diodes. Five levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type, and two levels of product assurance for each unencapsulated device type die as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (DO-7 and DO-35), figure 2 (DO-213AA), figure 3 (JANHCA and JANKCA), figure 4 (JANHCB and JANKCB), and figure 5 (JANHCC and JANKCC).

1.3 Unless otherwise specified, maximum ratings (T_A = +25°C).

P_T	T_{STG} and T_{op}	I_{ZM} 1/	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-55 to +175	70	3.33

1/ To guarantee voltage temperature stability, it is necessary to maintain the proper Iz as specified in paragraph 1.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 0.5.mA dc	V_Z I_Z = 0.5 mA dc		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4565A-1, 1N4565AUR-1	0.100	200	6.08	6.72	2.0
1N4566A-1, 1N4566AUR-1	0.050	"	"	"	"
1N4567A-1, 1N4567AUR-1	0.020	"	"	"	"
1N4568A-1, 1N4568AUR-1	0.010	"	"	"	"
1N4569A-1, 1N4569AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature Stability)	Z_Z I_Z = 1.0 mA dc	V_Z I_Z = 1.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4570A-1, 1N4570AUR-1	0.100	100	6.08	6.72	2.0
1N4571A-1, 1N4571AUR-1	0.050	"	"	"	"
1N4572A-1, 1N4572AUR-1	0.020	"	"	"	"
1N4573A-1, 1N4573AUR-1	0.010	"	"	"	"
1N4574A-1, 1N4574AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 2.0 mA dc	V_Z I_Z = 2.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4575A-1, 1N4575AUR-1	0.100	50	6.08	6.72	2.0
1N4576A-1, 1N4576AUR-1	0.050	"	"	"	"
1N4577A-1, 1N4577AUR-1	0.020	"	"	"	"
1N4578A-1, 1N4578AUR-1	0.010	"	"	"	"
1N4579A-1, 1N4579AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature Stability)	Z_Z I_Z = 4.0 mA dc	V_Z I_Z = 4.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4580A-1, 1N4580AUR-1	0.100	25	6.08	6.72	2.0
1N4581A-1, 1N4581AUR-1	0.050	"	"	"	"
1N4582A-1, 1N4582AUR-1	0.020	"	"	"	"
1N4583A-1, 1N4583AUR-1	0.010	"	"	"	"
1N4584A-1, 1N4584AUR-1	0.005	"	"	"	"

1/ Electrical characteristics and test conditions for " A-1, and AUR-1" devices are identical.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Performance Specification, Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be specified in MIL-PRF-19500 and figures 1, 2, 3, 4, and 5 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.4.2 Diode construction. These devices shall be constructed in a manner and using material which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.4.3 Dash-one construction. Shall be as specified in MIL-PRF-19500.

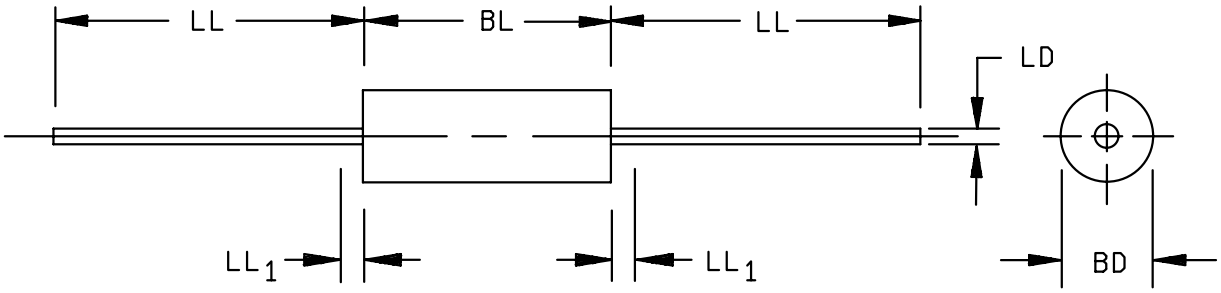
3.4.4 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-PRF-19500.

3.4.5 JANHC and JANKC construction. JANHC and JANKC construction may differ in die size and bonding pad layout provided the manufacturing technology is identical (example: Diffused junction, alloy junction).

3.4.6 Package outline. This specification contains two standard packages; DO-7 and DO-35. Any user of this specification that has a specific package outline requirement shall specify their preference in the document purchase order. If package is not specified, the manufacturer may supply either package (see 6.2).

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

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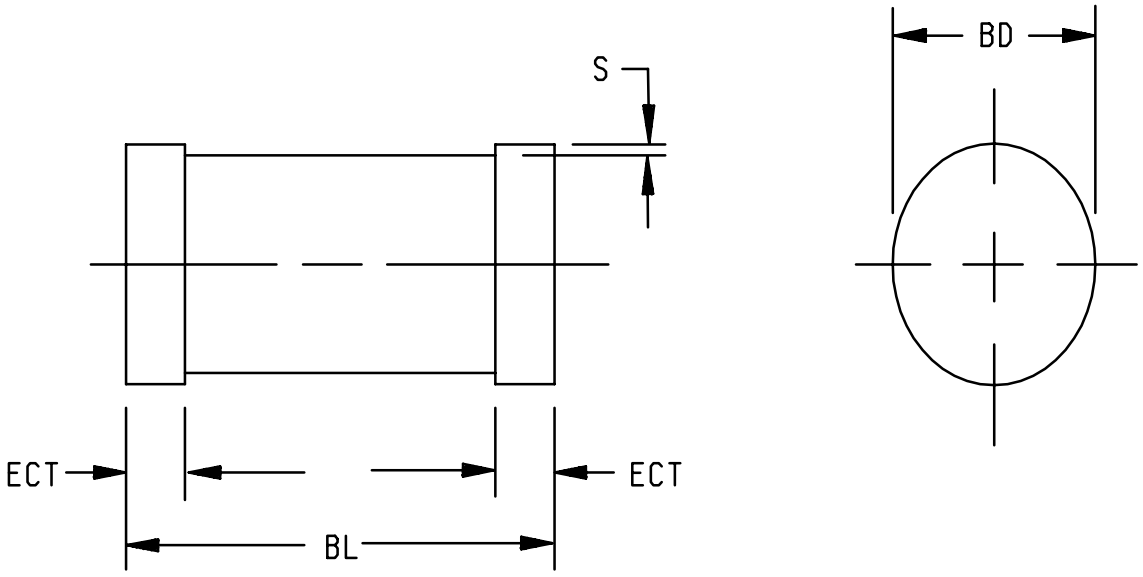


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	3
BL	.120	.300	3.05	7.62	3
LD	.018	.023	0.46	0.58	
LL	1.000	1.500	25.40	38.10	
LL_1	---	0.050	---	1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.

FIGURE 1. Physical dimensions 1N4565A-1 THROUGH 1N4584A-1,(DO-35 and DO-7).

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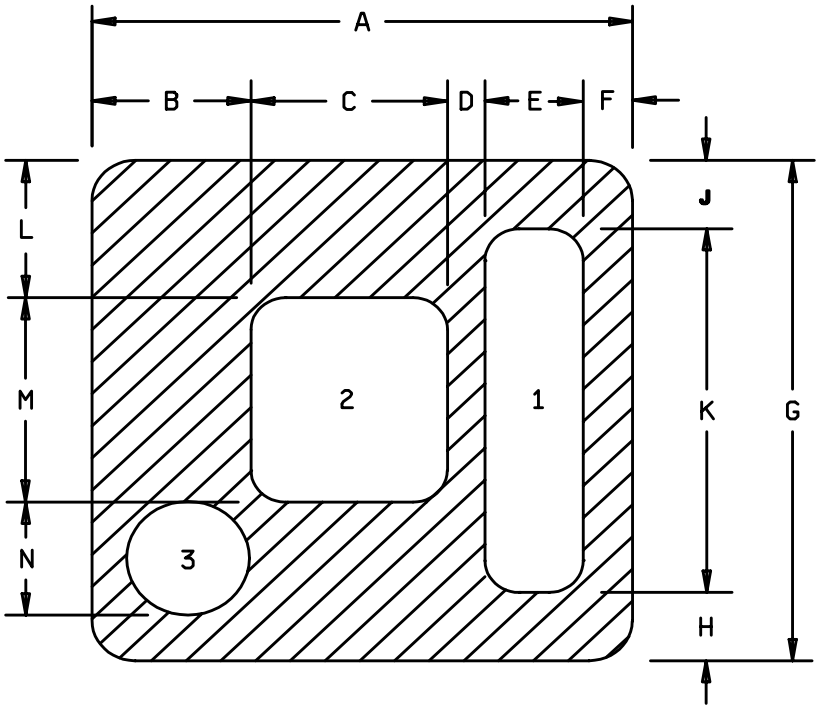
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
ECT	.016	.022	0.41	0.56
BL	.130	.146	3.30	3.71
S	.001 Min		0.03 Min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions 1N4565AUR-1 THROUGH 1N4584AUR-1 (DO-213AA).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0280	.0320	.711	.813
B	.0080	.0100	.203	.254
C	.0104	.0106	.264	.269
D	0019	.0021	.048	.053
E	.0054	.0056	.137	.142
F	.0020	.0040	.050	.102
G	.0280	.0320	.711	.813
H	.0030	.0050	.076	.127
J	.0030	.0050	.076	.127
K	.0209	.0211	.531	.536
L	.0080	.0100	.203	.254
M	.0104	.0106	.264	.269
N	.0059	.0061	.150	.155



BACKSIDE MUST BE ELECTRICALLY ISOLATED TO ENSURE PROPER PERFORMANCE

DESIGN DATA

Metallization:

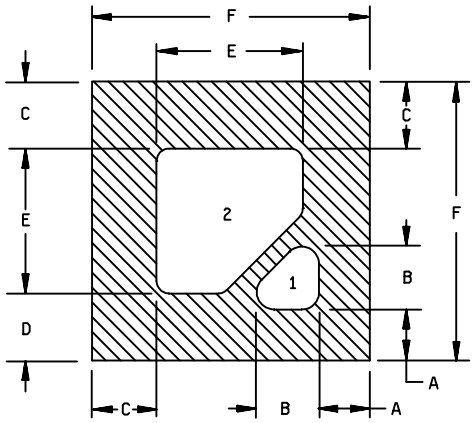
Top:	1 (Cathode)	Al	Circuit layout data:
	2 (Anode)	Al	For zener operation, cathode must be operated positive with respect to anode.
	3 (Test pad)	Al	Test pad is for wire bond evaluation only. No electrical contact is made with test pad.
Back:		Au	

Al thickness	25,000Å minimum
Gold thickness	4,000Å minimum
Chip thickness	0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches unless otherwise indicated.
2. Metric equivalents are given for general information only.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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BACKSIDE MUST BE ELECTRICALLY ISOLATED

Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0035	.0065	.088	.165
B	.0050	.0080	.127	.200
C	.0050	.0065	.127	.165
D	.0050	.0065	.127	.165
E	.0150	.0165	.381	.419
F	.0260	.0290	.660	.737

DESIGN DATA

Metallization:

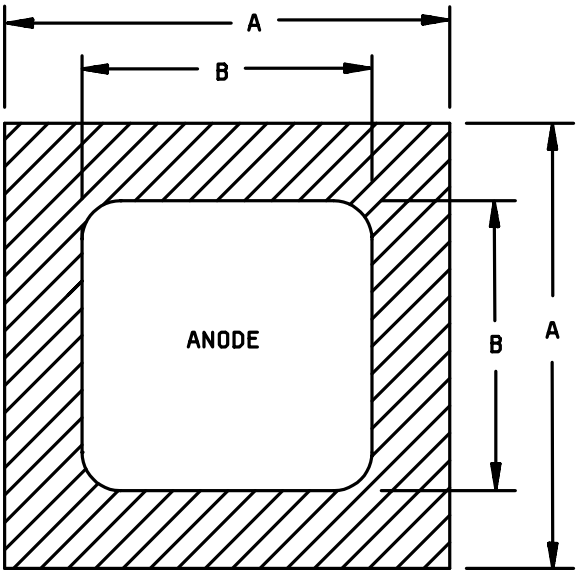
Circuit layout data:		2. (Anode)	Al
For zener operation, cathode must be	Top:	1. (Cathode)	Al
operated positive with respect to anode.	Back		Au

Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 4. JANHC and JANKC (B-version) die dimensions.

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BACKSIDE IS CATHODE

Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
A	.024	.037	0.61	0.94
B	.014	.029	0.36	0.74

DESIGN DATA

Metallization:

Circuit layout data For zener operation, cathode must be operated positive with respect anode.	Top:	(Anode)	Al
	Back		Au
	Al thickness		40,000Å minimum
	Gold thickness		5,000Å minimum
	Chip thickness		0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 5. JANHC and JANKC (C-version) die dimensions.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.3).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 6 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and paragraph 4.4.5 herein.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500.

4.2.3 JANJ devices. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or controls shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. When alternate screens, procedures, and/or controls are used, in lieu of the JANJ screens herein equivalency shall be proven and documented in the QM Plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, paragraph 3.3.1.1 of MIL-PRF-19500 is not required. Die lot control and rework shall be in accordance with MIL-PRF-19500 paragraph 3.13 and D 3.13.2.1 for JANS level. Lot formation and conformance inspection requirements for JANJ shall be those used for JANTXV devices as a minimum.

4.2.4 JANJ Qualification. For JANJ qualification, 4.4.2.1 herein shall be performed as required by the Qualifying Activity.

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4.3 Screening (JANS, JANJ, JANTXV and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	JANS Level	JANJ Level	JANTXV and JANTX Level
1a	Required	Not Required	Not Required
1b	Required	Required	Required (JANTXV only)
2	Not Required	Not Required	Not Required
3a	Required	Required	Required
3b	Not Applicable	Not Applicable	Not Applicable
3c	Not Applicable	Not Applicable	Not Applicable
4	Not Applicable	Not Applicable	Not Applicable
5	Not Applicable	Not Applicable	Not Applicable
6	Not Applicable	Not Applicable	Not Applicable
7a	Not Applicable	Not Applicable	Not Applicable
7b	Required	Required	Required
8	Required	Not Required	Not Required
9	Not Applicable	Not Applicable	Not Applicable
10	Not Applicable	Not Applicable	Not Applicable
11	Required V_Z, Z_Z	Required V_Z, Z_Z	Required V_Z, Z_Z
12	Required See 4.3.1	Required T = 240 hrs. See 4.3.1	Required See 4.3.1
13	Required Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $\Delta V_Z \leq \pm 0.005$ V dc from initial value at $T_A = +25°C \pm 2°C$	Required Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $\Delta V_Z \leq \pm 0.005$ V dc from initial value at $T_A = +25°C \pm 2°C$	Required Subgroup 2 of table I herein $\Delta Z_Z \leq \pm 15$ percent of initial reading. $\Delta V_Z \leq \pm 0.005$ V dc from initial value at $T_A = +25°C \pm 2°C$
14a	Not Applicable	Not Applicable	Not Applicable
14b	Optional 1/	Optional 1/	Optional 1/
15	Required	Required Attributes Data only Film or Non-Film Techniques may be utilized	Not Required
16	Required	Required	Not Required
17	Not Required	Required Subgroup 2 of table I herein Verify polarity	Not Required

1/ Per MIL-PRF-19500

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

1N4565 through 1N4569: I_Z = 0.5 mA dc ±0.05 mA dc, T_A = +150 +5°C, -0°C.
1N4570 through 1N4574: I_Z = 1.0 mA dc ±0.1 mA dc, T_A = +150 +5°C, -0°C.
1N4575 through 1N4579: I_Z = 2.0 mA dc ±0.2 mA dc, T_A = +150 +5°C, -0°C.
1N4580 through 1N4584: I_Z = 4.0 mA dc ±0.4 mA dc, T_A = +150 +5°C, -0°C.

To better utilize burn-in equipment, higher values of I_Z shall be permitted provided the junction temperature does not exceed +175°C.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, tables VIa (JANS) and VIb (JAN, JANTX , JANTXV, and JANJ) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, Subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for each die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest ΔV_Z rating in the inspection lot.

Subgroup	Method	Conditions
B1	2066	
B2	2026 1022	
B3	1056 4066 1071 2075	Test Condition A, 25 Cycles Not applicable Test condition E.
B4	1037	I_z = 35 mA dc at T_A = Room Ambient; t_{on} = t_{off} = 30 seconds minimum for 4,000 cycles.
B5	1027	I_{ZM} = 70 mA dc for 96 hours. T_A = +75°C or adjusted as required, to give an average lot T_J = +200°C.
B6		Not applicable

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX, JANTXV, and JANJ) of MIL-PRF-19500.

Subgroup	Method	Conditions
B1	2026 1022	
B2	1056 4066 1071	Test Condition A, 25 Cycles Not applicable Test condition E
B3	1027	Conditions for " A-1, and AUR-1" versions of: (See 4.3.1)
B4	2075	
B5		Not applicable
B6		Not applicable

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table I, Subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C1	2066	
C2	1056 2036 1071 1021	Lead fatigue: Test condition E; tension: test condition A; 4 pounds weight, t = 15 ±3 seconds; (lead fatigue and tension tests are not applicable to surface mount "UR" version devices). Omit initial conditioning
C3		Not applicable
C4	1041	
C5		Not applicable
C6	1026	Conditions for " A-1, and AUR-1" versions of: 1N4565 through 1N4569: I_Z = 0.5 mA dc, T_A = +100°C.(see 4.5.2) 1N4570 through 1N4574: I_Z = 1.0 mA dc, T_A = +100°C. (see 4.5.2) 1N4575 through 1N4579: I_Z = 2.0 mA dc, T_A = +100°C. (see 4.5.2) 1N4580 through 1N4584: I_Z = 4.0 mA dc, T_A = +100°C. (see 4.5.2)
C7		Not applicable

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500. MIL-PRF-19500, appendix F is not required when homogeneous product of radiation hard assembly (in accordance with the applicable 36D form) is submitted and successfully passes group D sampling of total dose at an outside facility compliant to MIL-STD-750, method 1019. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the submitted 36D forms.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table I, Subgroup 2 herein.

4.4.5.1 Group E inspection, appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Conditions	Sample Plan
E1	1051	500 cycles	22 devices, c = 0.
E2	1038	Condition B, 1,000 hours (see 4.3.1)	22 devices, c = 0.
E3		Not applicable	
E4		Not applicable	
E5		Not applicable	
E6		Not applicable	

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Voltage temperature stability. The breakdown voltage of each type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z per diode type.

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4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing, at 340 hours, and at the conclusion of the life test. The 340 hour reading shall be compared with the 0-hour reading and the 1,000 hour reading compared with the 340 hour reading. The change in breakdown voltage shall not exceed the limits specified. The test temperature for breakdown voltage shall be the same as the specified ambient life test temperature.

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Condition 1/		Min	Max	
1	Reference-voltage stability		T_A = +100°C ±2°C (0 to 340 hours see 4.5.2 and 4.5.3)	ΔV_Z			mV dc
	1N4565, 1N4570					10	
	1N4575, 1N4580					10	
	1N4566, 1N4571					8	
	1N4576, 1N4581					8	
	1N4567, 1N4572					7	
	1N4577, 1N4582					7	
	1N4568, 1N4573					6	
	1N4578, 1N4583					6	
	1N4569, 1N4574					5	
	1N4579, 1N4584					5	
2	Reference-voltage stability		T_A = +100°C ±2°C (340 to 1000 hours see 4.5.2 and 4.5.3)	ΔV_Z			mV dc
	1N4565, 1N4570					5	
	1N4575, 1N4580					5	
	1N4566, 1N4571					4	
	1N4576, 1N4581					4	
	1N4567, 1N4572					4	
	1N4577, 1N4582					4	
	1N4568, 1N4573					3	
	1N4578, 1N4583					3	
	1N4569, 1N4574					3	
	1N4579, 1N4584					3	

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 s maximum) prior to reading the reference voltage. For this test, the diode shall be suspended by its' leads with mounting clips whose inside edge is located at 0.375 inch (9.53 mm) from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provide thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical Examination	2071					
Subgroup 2						
Reference voltage	4022	(See 4.5.3)	V_Z	6.08	6.72	V dc
1N4565 through 1N4569 1N4570 through 1N4574 1N4575 through 1N4579 1N4580 through 1N4584		I_Z = 0.5 ±0.01 mA dc I_Z = 1.0 ±0.01 mA dc I_Z = 2.0 ±0.01 mA dc I_Z = 4.0 ±0.01 mA dc				
Small signal breakdown impedance	4051		Z_Z			ohms
1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc I_{sig} = 0.05 mA rms			200	
1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc I_{sig} = 0.1 mA rms			100	
1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc I_{sig} = 0.2 mA rms			50	
1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc I_{sig} = 0.4 mA rms			25	
Subgroup 3						
Voltage-temperature stability (see 4.5.1 and 4.5.3)		T_A = -55°C, 0°C, +25°C, +75°C, +100°C ±2°C, I_Z = (see 1.4)	ΔV_Z			mV dc
1N4565A-1, 1N4565AUR-1					100	
1N4570A-1, 1N4570AUR-1					100	
1N4575A-1, 1N4575AUR-1					100	
1N4580A-1, 1N4580AUR-1					100	
1N4566A-1, 1N4566AUR-1					50	
1N4571A-1, 1N4571AUR-1					50	
1N4576A-1, 1N4576AUR-1					50	
1N4581A-1, 1N4581AUR-1					50	

See footnote at end of table I.

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TABLE I. Group A inspection. -Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 – Continued						mV dc
1N4567A-1, 1N4567AUR-1					20	
1N4572A-1, 1N4572AUR-1					20	
1N4577A-1, 1N4577AUR-1					20	
1N4582A-1, 1N4582AUR-1					20	
1N4568A-1, 1N4568AUR-1					10	
1N4573A-1, 1N4573AUR-1					10	
1N4578A-1, 1N4578AUR-1					10	
1N4583A-1, 1N4583AUR-1					10	
1N4569A-1, 1N4569AUR-1					5	
1N4574A-1, 1N4574AUR-1					5	
1N4579A-1, 1N4579AUR-1					5	
1N4584A-1, 1N4584AUR-1					5	
Subgroups 4, 5, and 6						
Not applicable						
Subgroup 7						
Reverse Leakage Current	4016	DC Method; V_R = 3.0V	I_R		2.0	µA

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Pre-irradiation limits M, D, L, R, F, G, and H Min	Pre-irradiation limits M, D, L, R, F, G, and H Max	Post-irradiation limits M, D, L, R, F, G, and H Min	Post-irradiation limits M, D, L, R, F, G, and H Max	$\Delta V_{Z\theta}$ Pre-Post I_{RRAD} change Min	$\Delta V_{Z\theta}$ Pre-Post I_{RRAD} change Max	Unit
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = (see 1.4) condition A								
Reference voltage (see 4.5.3)	4022	I_Z = (see 1.4)	V_Z	6.08	6.72	6.08	6.72			V dc
Small-signal breakdown impedance	4051	I_Z = (see 1.4)	Z_Z							Ω
1N4565A		I_{sig} = 0.05 mA ac			200		200			
1N4566A		I_{sig} = 0.05 mA ac			200		200			
1N4567A		I_{sig} = 0.05 mA ac			200		200			
1N4568A		I_{sig} = 0.05 mA ac			200		200			
1N4569A		I_{sig} = 0.05 mA ac			200		200			
1N4570A										
1N4571A		I_{sig} = 0.1 mA ac			100		100			
1N4572A		I_{sig} = 0.1 mA ac			100		100			
1N4573A		I_{sig} = 0.1 mA ac			100		100			
1N4574A		I_{sig} = 0.1 mA ac			100		100			
		I_{sig} = 0.1 mA ac			100		100			
1N4575A										
1N4576A		I_{sig} = 0.2 mA ac			50		50			
1N4577A		I_{sig} = 0.2 mA ac			50		50			
1N4578A		I_{sig} = 0.2 mA ac			50		50			
1N4579A		I_{sig} = 0.2 mA ac			50		50			
		I_{sig} = 0.2 mA ac			50		50			
1N4580A										
1N4581A		I_{sig} = 0.4 mA ac			25		25			
1N4582A		I_{sig} = 0.4 mA ac			25		25			
1N4583A		I_{sig} = 0.4 mA ac			25		25			
1N4584A		I_{sig} = 0.4 mA ac			25		25			
		I_{sig} = 0.4 mA ac			25		25			

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		$\Delta V_{Z\theta}$ Pre-Post I_{RRAD} change		Unit
	Method	Conditions		M, D, L, R, F, G, and H		M, D, L, R, F, G, and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 2 - Cont										
Reverse current leakage	4016	DC method; V_R = 3.0 V dc	I_R		2.0		2.0			µA
Voltage stability (see 4.5.3)	4022	I_Z = (see 1.4) T_A = +25°C ±2°C	ΔV_Z							mV
1N4565A									± 3	
1N4566A									± 3	
1N4567A									± 2	
1N4568A									±1.5	
1N4569A									± 1	
1N4570A									± 3	
1N4571A									± 3	
1N4572A									± 2	
1N4573A									±1.5	
1N4574A									± 1	
1N4575A									± 3	
1N4576A									± 3	
1N4577A									± 2	
1N4578A									±1.5	
1N4579A									± 1	
1N4580A									± 3	
1N4581A									± 3	
1N4582A									± 2	
1N4583A									±1.5	
1N4584A									± 1	

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics and test conditions for " A-1, and AUR-1" devices are identical.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCAM1N4565A) will be identified on the parts list of MIL-PRF-19500.

JANC ordering information					
PIN	Manufacturer				
	55801	55801 Radiation designators M,D,L,R,F,G,H	12954	12954 Radiation designators M,D,L,R,F,G,H	12954
1N4565A through 1N4584A	JANHCA1N4565A through JANHCA1N4584A	JANHCA1N4565A through JANHCA1N4584A	JANHCB1N4565A through JANHCB1N4584A	JANHCB1N4565A through JANHCB1N4584A	JANHCC1N4565A through JANHCC1N4584A
1N4565A through 1N4584A	JANKCA1N4565A through JANKCA1N4584A	JANKCA1N4565A through JANKCA1N4584A	JANKCB1N4565A through JANKCB1N4584A	JANKCB1N4565A through JANKCB1N4584A	JANKCC1N4565A through JANKCC1N4584A

6.5 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.6 Substitution of devices. ΔV_Z Device types within this series with higher type numbers (lower ΔV_Z) are a direct oneway substitution for lower type numbers (higher ΔV_Z).

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA – CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/452E	**2. DOCUMENT DATE (YYYYMMDD)**

DOCUMENT TITLE Semiconductor device, Diode, Silicon, Low Level Voltage-Reference Types 1N4565A-1 through 1N4584A-1 and 1N4565AUR-1 through 1N4584AUR-1 JAN, JANTX, JANTXV, JANJ, JANS, JANHC, JANKC; Radiation Hardened (Total Does Only) Types JANTXVM D, L, R, F, G, H; JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; AND JANKCM, D, L, R, F, G, H

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)* *(3) E-mail*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE *Include Area Code)* (1) Commercial (614)692-0510 (2) DSN 850-0510 (3) E-mail alan_barone@dscc.dla.mil
c. ADDRESS *(Include Zip Code)* Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE STANDARDIZATION PROGRAM OFFICE (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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